ECE 501 Project Week 6 Report

Project Title: Vehicle Orientation Estimation Using Conventional Image Processing

Group Number: 14

Project number: 8

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Github Repository Link: https://github.com/hetpatel5107/DIP_2026_Project8_Group14

- **Introduction**

This week, we extended our vehicle orientation estimation experiment by applying the Canny edge detection and Radon transform method to five aerial images from the dataset. Using the provided annotations, we extracted individual vehicle crops and compared the estimated orientation axes with the ground-truth angles. We then evaluated the results using overall error statistics, error distribution, and comparisons between vehicle types.

- **Error Statistics and Error for each vehicle type**

Code 1:

```
results["Radon Error"] = abs(results["Radon Estimate"] - results["Ground
Truth (0-90)"])
valid = results.dropna(subset=["Radon Estimate"])

print("Crops with an estimate:", len(valid), "of", len(results))
print("Mean Absolute Error:", round(valid["Radon Error"].mean(), 2),
"degrees")
print("Median Error:", round(valid["Radon Error"].median(), 2), "degrees")
print("Minimum Error:", round(valid["Radon Error"].min(), 2), "degrees")
print("Maximum Error:", round(valid["Radon Error"].max(), 2), "degrees")
print("Within 10 degrees:", round((valid["Radon Error"] <= 10).mean() *
100, 1), "%")
```

Code 2:

```
by_type = valid.groupby("Type")["Radon Error"].agg(
    Count="size", MAE="mean", Median="median"
).sort_values("Count", ascending=False).round(2)

print(by_type.to_string())

ax = by_type["MAE"].plot(kind="bar", figsize=(10, 5))
for i, (n, mae) in enumerate(zip(by_type["Count"], by_type["MAE"])):
    ax.text(i, mae + 0.5, "n=" + str(n), ha="center")
plt.ylim(0, by_type["MAE"].max() * 1.2)
plt.xlabel("Vehicle Type")
plt.ylabel("Mean Absolute Error (degrees)")
plt.title("Radon Orientation Error by Vehicle Type")
plt.xticks(rotation=35, ha="right")
plt.tight_layout()
plt.show()
```

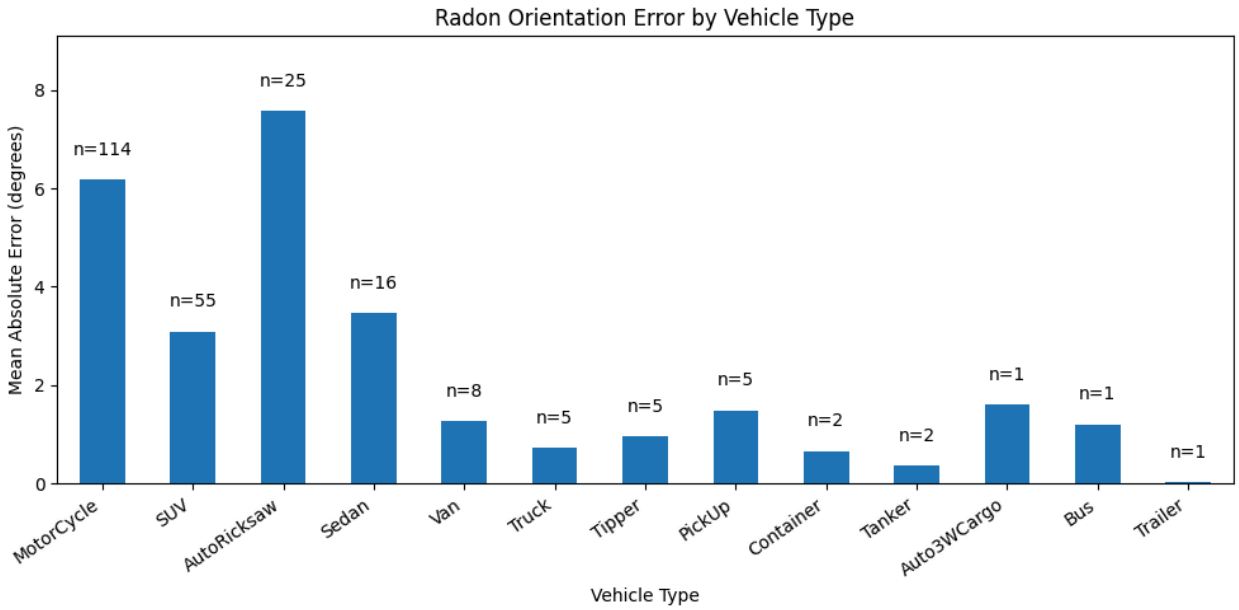
Output 1:

Crops with an estimate: 240 of 240
Mean Absolute Error: 4.79 degrees
Median Error: 2.6 degrees
Minimum Error: 0.0 degrees
Maximum Error: 80.0 degrees
Within 10 degrees: 90.8 %

Output 2:

Type	Count	MAE	Median
MotorCycle	114	6.18	4.42
SUV	55	3.09	2.10
AutoRicksaw	25	7.58	2.43
Sedan	16	3.47	2.60
Van	8	1.26	0.40
Truck	5	0.73	0.62
Tipper	5	0.95	0.50
PickUp	5	1.49	1.04
Container	2	0.66	0.66
Tanker	2	0.36	0.36
Auto3WCargo	1	1.60	1.60
Bus	1	1.20	1.20
Trailer	1	0.02	0.02

Plot:



Interpretation:

The first glance at the statistics is very promising. In all 240 vehicle crops, the Canny and Radon combination performed with a Mean Absolute Error (MAE) of only 4.79 ° and a median error of 2.6 °. Most importantly, 90% of our estimates were within 10 degrees of the ground truth. This is a pretty decent starting point for a traditional image processing workflow that doesn't depend on on-the-fly machine learning. But on closer examination of the bar chart and the comparison by vehicle type, there is a huge disparity in the errors. The algorithm is extremely precise on standard and heavy vehicles. SUVs (3.09° error), Sedans (3.47° error) and Vans (1.26° error) all had a remarkably high score. Average errors for large vehicles, such as Trucks and Tippers, even decreased to less than one degree. Conversely, smaller and non-standard shaped vehicles were finding it difficult. Motorcycles and Auto-rickshaws had the highest average errors at 6.18° and 7.58° respectively. Unfortunately, these two vehicle types accounted for the majority of our dataset (139 out of 240 crops), so they pulled our averages down. There's a reason for this split in the pipeline: The Radon transform is looking for the longest and strongest straight line in the image so as to define the

direction of the vehicle! In a way, cars, vans and SUVs are just large boxes with geometry. They have long, straight and continuous side panels and roofs, viewed from a drone's top-down look. Canny can easily outline them nicely in a rectangular style.Rickshaws and motorcycles, however, are uneven and bumpy in shape, open to the rider, and don't have long straight edges. This makes the Canny filter detect a chaotic and broken-out outline rather than a neat box. If we pass that noisy outline into the Radon transform, due to the lack of a dominant, distinctive straight line to which the math can be secured, the heading error will be much higher.

- **Error Distribution**

Code:

```
plt.figure(figsize=(8, 5))



plt.hist(valid["Radon Error"], bins=18, range=(0, 90), edgecolor="black")

plt.xlabel("Absolute Error (degrees)")

plt.ylabel("Number of Vehicles")

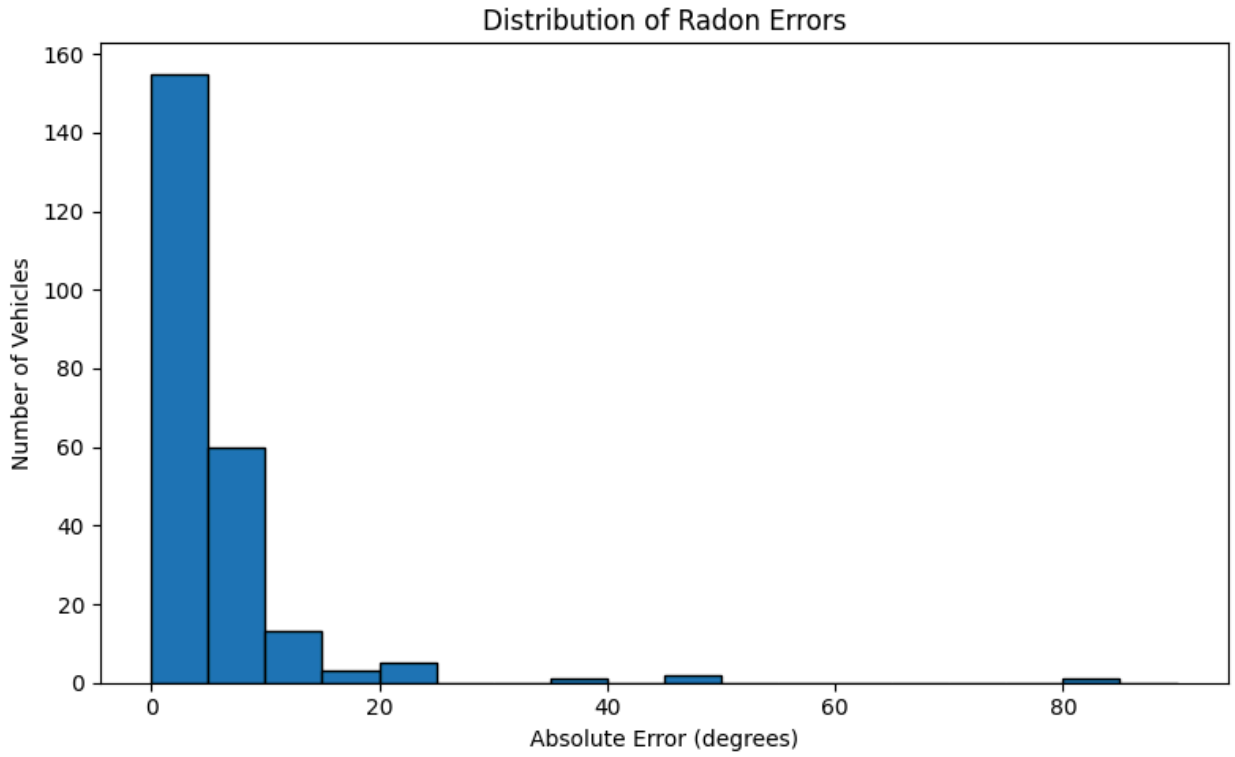
plt.title("Distribution of Radon Errors")



plt.tight_layout()

plt.show()
```

Output:



Interpretation:

The histogram shows that most vehicles have relatively low Radon orientation errors, with most of them between 0° and 5°. The number of vehicles decreases as the error increases, indicating that the method estimates the orientation accurately for many crops. However, a small number of vehicles have much larger errors, including some above 20° and a few around 40° to 85°. These outliers show that the method does not perform equally well on every vehicle crop. Background edges, shadows, or vehicle shapes may contribute to these errors. Also as seen from error for each vehicle type plot, some vehicle types show larger errors giving these outliers. Overall, the distribution indicates good performance on many crops, alongside a small number of significant failures.

- **CSV Results**

Code:

```
results.round(2).to_csv("week6_radon_results.csv", index=False)

by_type.to_csv("week6_radon_error_by_type.csv")
```

Output:

Type	Count	MAE	Median
MotorCycle	114	6.18	4.42
SUV	55	3.09	2.1
AutoRicksa	25	7.58	2.43
Sedan	16	3.47	2.6
Van	8	1.26	0.4
Truck	5	0.73	0.62
Tipper	5	0.95	0.5
PickUp	5	1.49	1.04
Container	2	0.66	0.66
Tanker	2	0.36	0.36
Auto3WCa	1	1.6	1.6
Bus	1	1.2	1.2
Trailer	1	0.02	0.02

Output Link:

https://drive.google.com/drive/folders/10EDej_vlq7kELK1AidALQNvdDbidD44W?usp=sharing

Interpretation:

The vehicle-level CSV contains 240 vehicle records from five aerial images. The Radon-based method achieved a mean absolute orientation error of approximately 4.79° and a median error of 2.60°. A total of 218 vehicles (90.83%) had errors within

10°, indicating that the method produced reasonably close estimates for most vehicles. However, eight vehicles had errors greater than 20°, showing that some estimates were substantially less accurate.

The vehicle-type CSV shows differences in performance across categories. Motorcycles had the largest number of samples (114) and a mean absolute error of 6.18°. Auto-rickshaws had the highest mean error, at 7.58°, among the listed categories. Their median error was 2.43°, which is much lower than their mean, suggesting that a small number of large errors may be increasing the average. In comparison, SUVs had 55 samples and a mean error of 3.09°. Results for the category with one or two vehicles need to be taken with caution because of very small sample sizes, which do not allow generalizations.

These discrepancies could be associated with the edges available for each car. Canny finds strong intensity boundaries, and Radon chooses the prevailing direction out of these edges. Shadows, background edges, shape of the car, or even cropping of the image can influence the prevailing edge direction choice. The information is contained in the CSV output file results; however, it does not necessarily show the cause of this particular mistake.

In general, the results show that Canny/Radon estimates the direction within ±10° for the majority of vehicles used in this experiment. However, the performance of the method varies across various vehicles and there are some errors. The analysis of high-error crops would help to determine the reason for mistakes made.

- **Conclusion**

This week, we extended the Canny–Radon method to a larger set of 240 vehicle crops from five aerial images. We evaluated the orientation estimates using error statistics, an error distribution histogram, and a comparison of errors across vehicle types. The results provide a broader view of how the method performs on different vehicles and image conditions. The individual predictions and summary results were also saved as CSV files for further analysis.

1. Work done this week

- Applied the Canny edge detection and Radon transform method to 240 vehicle crops extracted from five aerial images.
- Calculated the ground-truth orientation for comparison using the required 0°–90° range.
- Evaluated the predictions using overall error statistics and an error distribution histogram.
- Analysed orientation estimation errors across different vehicle types.
- Saved the individual prediction results and vehicle-type error summaries as CSV files for further analysis.

2. Work to be done next week

- Implement PCA-based vehicle orientation estimation, beginning with a single vehicle crop.
- Apply the PCA method to the same 240 vehicle crops used for the Radon method.
- Calculate and analyse the PCA orientation errors using the same evaluation metrics.

3. Challenges faced

- The Radon method estimates the dominant axis but cannot independently distinguish the front of a vehicle from its rear.
- Evaluating the method across multiple vehicles revealed the need to examine performance across different vehicle types and image conditions, rather than relying on a single example.